Triggers in MySQL – Exercise 19

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1. CREATE THE EMPLOYEES TABLE AND INSERT RECORDS

   DROP TABLE IF EXISTS employees;

   CREATE TABLE employees (id INT PRIMARY KEY, name VARCHAR(100), email VARCHAR(100));

   INSERT INTO employees VALUES

   (1, 'Harry Potter', 'pharry@warnerbros.com'),

   (2, 'Clark Kent', 'kclark@dccomics.com'),

   (3, 'Tony Stark', 'stony@marvel.com');

   Output:

```
Query OK, 0 rows affected (0.077 sec)

Query OK, 0 rows affected (0.110 sec)

Query OK, 3 rows affected (0.040 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

2. CREATE THE AUDIT TABLE EMPLOYEES_AUDIT

```
CREATE TABLE employees_audit (
    audit_id  INT AUTO_INCREMENT PRIMARY KEY,
    id        INT NOT NULL,
    name      VARCHAR(100),
    old_email VARCHAR(100),
    new_email VARCHAR(100),
    changedat DATETIME DEFAULT NULL
);
```

   Output:

```
Query OK, 0 rows affected (0.041 sec)
```

3. CREATE A BEFORE UPDATE TRIGGER

```
DELIMITER $$
CREATE TRIGGER before_employee_update
BEFORE UPDATE ON employees
FOR EACH ROW
BEGIN
    INSERT INTO employees_audit
    SET id        = OLD.id,
        name      = OLD.name,
        old_email = OLD.email,
        new_email = NEW.email,
        changedat = NOW();
END$$
DELIMITER ;
```

Output:

```
Query OK, 0 rows affected (0.014 sec)
```

4. UPDATE A RECORD (fires the trigger)

```
UPDATE employees SET email = 'harry@warnerbros.com' WHERE id = 1;
```

Output:

```
Query OK, 1 row affected (0.008 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

5. UPDATE ANOTHER RECORD

```
UPDATE employees SET email = 'tony@marvel.com' WHERE id = 3;
```

Output:

```
Query OK, 1 row affected (0.016 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

6. VIEW THE AUDIT TABLE

```
SELECT * FROM employees_audit;
```

Output:

```
+----------+----+--------------+----------------------+----------------------+---------------------+
| audit_id | id | name         | old_email            | new_email            | changedat           |
+----------+----+--------------+----------------------+----------------------+---------------------+
|        1 |  1 | Harry Potter | pharry@warnerbros.com | harry@warnerbros.com | 2026-10-28 14:41:07 |
|        2 |  3 | Tony Stark   | stony@marvel.com      | tony@marvel.com      | 2026-10-28 14:41:52 |
+----------+----+--------------+----------------------+----------------------+---------------------+
2 rows in set (0.002 sec)
```

7. VIEW THE EMPLOYEES TABLE AFTER THE UPDATES

   SELECT * FROM employees;

   Output:

```
+----+--------------+----------------------+
| id | name         | email                |
+----+--------------+----------------------+
|  1 | Harry Potter | harry@warnerbros.com |
|  2 | Clark Kent   | kclark@dccomics.com  |
|  3 | Tony Stark   | tony@marvel.com      |
+----+--------------+----------------------+
3 rows in set (0.002 sec)
```

8. DROP THE TRIGGER

   DROP TRIGGER before_employee_update;

   Output:

```
Query OK, 0 rows affected (0.045 sec)
```